LAYOFFS · METHOD REVIEW · 2 SEPTEMBER 2026

What the SDID literature actually does

Twelve marketing papers and three replication codebases, read for empirical mechanics only. The short version: your architecture is the field's architecture, your inference is ahead of the published bar, and you are missing four cheap things that every one of these papers has.

12 PAPERS 3 REPLICATION REPOS READ LINE BY LINE 5 USE SDID AS THE PRIMARY ESTIMATOR

I The consensus architecture

Five of the twelve use SDID as the headline estimator. Every one of those five does the same three things: fit SDID separately per cohort or per treated unit, aggregate with treated-count weights, and get the standard error from a jackknife. Your pipeline does all three.

PAPER	DESIGN	FITS	AGGREGATION	SE	ZEROS	PRE-FI
Berman & Israeli 2022 MkSc	Staggered, 18 cohorts	18	$N_{treated}$ weights	JACKKNIFE placebo if 1 treated	log, not discussed	Pre-pe study p=.10
Guyt, van Lin & Keller 2025 MkSc	Staggered, 5 cohorts	5	$N_{treated}$, Clarke Alg. 3	JACKKNIFE vs bootstrap	levels; log as robustness	Correl .64→.3
Keller & Guyt 2026 JM	Single event, per-product	220	$N_{treated\ pairs}$	JACKKNIFE vs bootstrap	assortment rule, no transform	Correl .64→.7 placeb
Pachali & Datta 2025 MkSc	Staggered, per-date batches	107	$N_{treated}$ weights	PLACEBO 200 reps	log1p + IHS robustness	MSPE .0000

van Lin, Keller & Guyt 2025 WP	Single event	~98	never aggregated	JACKKNIFE	log1p + arcsin + rescale	Correl .58→.'
Lambrecht et al. 2024 JMR	Field test	1	—	JACKKNIFE + Newey–West	ln(x+1)	Angris test
Qian & Xie 2026 JMR	Single event, 1 treated	1	—	ROBUST / CLUSTERED	zeros in levels; log1p for a second DV	Dynar
Han, Yang & Simonov 2025	Single event, pooled	1	—	CLUSTERED	log(x+1), undefended	Sup-*t*
Wu & Ham 2026 JM	TWFE; SDID as check	1	—	2-WAY CLUSTER	log(1+y), levels alongside	CS lea placeb
Pape & Rossi 2026 MkSc	TWFE; SDID as check	1	—	CLUSTER BY COUNTRY	zeros→1, tested at 0.5 and dropped	Full le every
Li 2024 MkSc	Method: Forward DiD	—	—	asymptotic normal	—	R^2 *and* PMSE
Cheng et al. 2026 MkSc	Method: Deep DiD	—	—	none — stated limitation	—	Valida

Highlighted rows are the two designs closest to yours: many separate SDID fits, aggregated. Keller & Guyt run 220; Pachali & Datta run 107. You run ~1,700. That is a difference of degree, not of kind.

THE RULE FOR CHOOSING THE VARIANCE ESTIMATOR, STATED ONCE IN THE LITERATURE

Berman & Israeli are the only authors who write it down: **jackknife by default, placebo variance when a cohort has only one treated unit.** It is also enforced in the package itself — `synthdid`'s `jackknife_se()` silently returns `placebo_se()` when `N0 ==`

`nrow(Y) - 1`. Pachali & Datta use placebo throughout because their batches average ~5 treated playlists; with that few units the jackknife is inconsistent. **At firm level you have ~190 treated units and at game level ~1,700, so the jackknife is the correct choice and the placebo variance is not needed.**

II Where you are already ahead of the published bar

This is the part that should settle the "am I overcomplicating this" question. Five things you have built that *none* of these twelve papers reports.

KEEP

Clustering at the level of treatment assignment

In all five SDID papers, clustering appears *only* in the TWFE comparison arm — never in the SDID arm. Guyt jackknifes over households when treatment is assigned to municipalities. van Lin jackknifes over households when treatment is assigned to *one province*. Your delete-one-firm jackknife is the correct generalisation of Clarke et al.'s Algorithm 3 to a design where one announcement hits many titles, and no published SDID paper in this set does it.

`van Lin et al. 2025` — "standard errors are clustered at the household level (Cameron and Miller 2015)", with 1 treated province. `Guyt et al. 2025` — jackknife over 892 treated households in 7 municipalities.

KEEP

Verified coverage of your own standard error

Not one of the twelve runs a simulation to check that its reported interval covers. You did: realised size `0.049` against a nominal `0.05`, with the iid comparison caught at `0.110`. That is one appendix table and one sentence in the method section, and it is genuinely unusual.

Nearest equivalent in the set: `Guyt et al.` check jackknife against bootstrap and report "virtually similar"; `Keller & Guyt` the same, "virtually identical". Neither is a coverage check.

KEEP

Equivalence testing on the design battery

Every robustness section in this literature reports "not significantly different"

and moves on. Two papers do better informally — Guyt with a minimum detectable effect (€1.09 against an estimate of €0.40) and Pape & Rossi with a probability bound ("we can exclude a decline of more than 5% with probability 95.9%"). **Nobody runs a formal two-one-sided test.** Your bounds — the ridge cannot move the answer by more than 0.026, the donor rule by more than 0.011 — are a stronger claim than anything in the set.

Guyt et al. 2025 App. F; Pape & Rossi 2026 §5.2.

KEEP

Multiplicity accounting across the robustness family

Guyt et al. do this by hand in a footnote — "9 of 212 estimates (4.2%) attain significance at $p < .05$ … ignoring the SC method, 4 of 203 (2.0%)" — and it is the only multiplicity treatment in the twelve. Your family test plus BH does the same job formally.

Guyt et al. 2025, footnote 29.

KEEP

A placebo battery built as an experiment rather than a plot

Pachali & Datta's `build_placebo()` is the closest analogue: assign fake treatment dates to never-treated units, matching the number of treated units per date. That is your `diag_placebo.R`. They use it for the variance; you use it to measure bias, validate coverage, and decompose variance. Your version is a superset.

Pachali & Datta 2025, `synthetic_did.R` lines 73–99.

III What to cut, and what I got wrong

Four things I pushed you toward that this literature does not do. In three of them I was over-engineering; in one I was simply wrong about the convention.

DEMOTE

Randomization inference as the headline *p*-value

I recommended replacing the bias correction with a permutation test against the placebo distribution. **Not one of the twelve papers uses randomization**

inference for the treatment effect. Keller & Guyt's in-time placebo is a falsification test on pre-trends, not a variance estimator. Report the firm-clustered jackknife as the headline, exactly as the field does, and keep the RI *p*-value as one appendix row — where it is a genuine strength rather than an unfamiliar main-table object a referee has to be taught to read.

DEMOTE

Wild cluster bootstrap as the reported test

Also unused across all twelve, and none of them acknowledges the few-effective-cluster problem at all — including van Lin et al., who have *one* treated province. Your `G_eff ≈ 31` is real and worth a footnote with the bootstrap *p* beside the normal one. It is not worth the main table.

DROP

PPML as a co-primary estimator for the zeros problem

I over-called this. Every paper here with zeros uses `log(1+y)` and defends it with a robustness check, not a second estimator. The field's answer is three lines: log1p headline, **IHS as a robustness**, and a **rescaling check** — van Lin et al. re-run in eurocents rather than euros, which is precisely the Chen–Roth scale-dependence test done in one line. Pape & Rossi go furthest and it is the model to copy: replace zeros with 1, then with 0.5, then drop them entirely — and they disclose honestly that dropping zeros kills one of their secondary findings.

`van Lin et al. fn. 13 — "results are substantively similar … when the variables are expressed in eurocents (cf. euros), and when we use an arcsin transformation". Pachali & Datta Table G2 — IHS "very close". Pape & Rossi Table B7.`

REVERSE

Firm-equal weighting as the default aggregation

I had this backwards. **The convention is treated-count weighting** — Arkhangelsky et al. p. 4114, and all four aggregating papers implement it: Berman's `colSums(TE*Ntr/sum(Ntr))`, Guyt's `sum(ATE*N_treated)/sum(N_treated)`, Keller & Guyt "weighting the UPCs by the respective number of treated UPC–store pairs". With one SDID per game, treated-count weighting *is* game-equal weighting. So game weighting is the literature default, not the deviation.

Your 139-title firm is still a real concentration problem that none of these designs faces. The resolution is to report both in one table and lead with the game-weighted number for comparability — not to make the unconventional

choice the headline.

AND ONE THING YOU BUILT THAT THE LITERATURE INDEPENDENTLY CONFIRMS

Your ζ result — lowering the ridge raises median pre-fit R^2 from `0.35` to `0.49` while the placebo bias does not move — is exactly Li (2024)'s central warning, arrived at from the other direction. On his data, Forward HCW reaches "R^2 values greater than 0.99 for all three cities, which is another indication of overfitting," and the confidence intervals collapse to `[45,207.9, 45,208.1]`: "we cannot interpret these narrow CIs as more precise estimates but rather as the method breaking down." You rediscovered this by measurement. Cite him and the finding stops being a curiosity and becomes a contribution.

IV What to add — four cheap things, one substantial one

ADD

The raw-versus-weighted pre-treatment correlation table

Three of the five SDID papers use this as their *main* fit exhibit, and you do not have it. For each outcome: the pre-period correlation between the treated series and the SDID-weighted control, next to the correlation between the treated series and the unweighted control average. It answers "did the weighting buy anything?" in a form referees already know how to read, and it is about ten lines of code against your existing per-unit output.

van Lin et al.'s footnote 15 is worth stealing wholesale: where the *unweighted* correlation is already high, "a traditional DID model is unbiased and more efficient (see Li and Van den Bulte 2023)." That is the justification for reporting your `estimator = "did"` arm alongside — which you already have and have never used.

Guyt et al. raw .64 → SDID .86. **Keller & Guyt** .64 → .70 sales, .75 → .84 prices. **van Lin et al.** .58 → .71. Implementation: `02_analysis.R` lines 145–166.

ADD

The backdating test — the one diagnostic R^2 cannot give you

Split the pre-period at some $T_1 < T_0$, fit τ and λ on weeks $1 \dots T_1$, predict

Split the pre-period at some $T_0' < T_0$, fit ω and λ on weeks $1...T_0'$, predict weeks $T_0'+1...T_0$, and score by out-of-sample prediction error. This is the only check in the whole literature that tests whether the fitted weights *generalise* rather than whether they fit. Li reports 43 of 60 PMSE ratios above one for his method against the alternatives.

For you it does two jobs at once: it is the honest test of whether your donor pool is too large for `T₀ = 52`, and it gives you a per-unit screen that is not selection-on-outcome — because it is computed entirely inside the pre-period. If it turns out that units with poor out-of-sample pre-fit are the ones carrying `|att| > 1`, that is a defensible sample restriction where an R^2 floor was not.

Li 2024 MkSc 43(2), backdating exercise with $T_0 \in \{12,14,16,18,20\}$.

ADD

Decide how subgroup weights are handled — it is an unmade decision

All three of the Guyt-adjacent papers confront the same fork when estimating subgroup effects, and they split on it. Berman & Israeli and Guyt et al. **hold $\hat{\omega}$ and $\hat{\lambda}$ fixed from the full sample** and swap in only the subgroup as treated, "so all units have the same synthetic control" — subgroups are then directly comparable, at the cost of subgroup-specific fit. van Lin et al. **re-estimate within subgroup**, because "our primary interest is to understand the effect for different subgroups."

Your heterogeneity layer needs this decided and stated. For moderator analysis where the comparison *between* subgroups is the point, hold the weights fixed.

Berman & Israeli endnote 34 — approach 1 "yielded similar but noisier results". **Guyt et al.** fn. 22. **van Lin et al.** fn. 16.

ADD

Weight the moderator regression by 1/SE — your per-unit estimates are a generated regressand

This is the substantial one, and it is the exact template for the heterogeneity layer you have been planning. Keller & Guyt take their 220 per-UPC SDID estimates and use them as the *dependent variable* in a second-stage regression on covariates, then estimate it "by OLS *and* WLS using the inverse of the first-stage SE as the weight, to account for the uncertainty in the sales change."

Your per-unit ATTs have placebo RMSE around `0.59` at firm level, and the

noise is *not* homoskedastic — it varies systematically with donor-pool size and pre-fit, both of which correlate with title size. An unweighted moderator regression on those estimates gives every game equal say regardless of how badly measured it is. This is a real bias, not just an efficiency loss, and there is a published marketing-journal precedent for the fix.

`Keller & Guyt 2026 JM Eq. 3 and Eq. 5 — 413 market-package-size effects and 99 retailer-package-size cells as dependent variables.`

ADD

Report SDID, DiD and SC side by side — you already have the switch

Four of the twelve report two or three estimators in the same table as a matter of course: Berman (SynthDiD / staggered DiD / IV), Wu & Ham (TWFE / CS / SDID / PSM-DiD), Pape & Rossi (TWFE / SDID / SC / triple-diff), Han et al. (TWFE / SDID / CS). Your `estimator = "sdid" | "did" | "sc"` runs the identical sample, screen, donors and diagnostics — so the comparison is free, and its whole point is exactly what your config comment already says: "if `did` returns the same ATT with a worse pre-fit, the weights are buying precision; if it returns a different ATT, they are buying identification."

Two smaller ones worth a line each

- **Residualise, don't impute.** Qian & Xie and Lambrecht et al. both handle seasonality and promotions by regressing the outcome on the controls first and running SDID on the *residuals*. That is cleaner than your `marker_mode = "impute"` and it is what two of these papers do. Worth adding as a third mode.
- **Count your in-time placebo failures and drop them.** Keller & Guyt run a fake treatment at the pre-period midpoint across all 220 fits, report "significant deviations for only a few effects (18 out of the 220 estimates)", and then show results are robust to dropping those UPCs. You compute `placebo_att` per unit already — this turns it